Interactive Learning Tool for Dyslexic Children

R26-IT-130

Project Proposal Report

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January 2026

1. DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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2. ABSTRACT

Background: Dyslexia is a neurological learning disorder affecting reading, spelling, and cognitive processing, with an estimated prevalence of 5% to 10% in most populations. In Sri Lanka, primary school teachers often lack the tools for early identification and intervention, leading to significant social and economic impacts such as low self-esteem and high school dropout rates.
Identified Research Gap: Existing international solutions are designed for English-speaking populations and lack cultural adaptation. Local tools like ARUNALU and LEXIS focus primarily on literacy practice or Sinhala-only modules, neglecting the bilingual reality of Sri Lankan education and the underlying executive function deficits like brain processing speed and visual tracking.
Proposed Solution Overview: This project introduces a bilingual (Sinhala-English), web-based gamified platform that integrates cognitive training with literacy intervention. My specific component focuses on a **Combined Eye-Tracking and Brain-Speed Assessment**. Unlike traditional tools that only measure reading outcomes, this system captures the real-time cognitive processes by recording gaze behavior such as fixation duration and regressions—during reading tasks.
Technologies to be used: The system utilizes standard web cameras for low-cost eye-tracking, avoiding expensive medical hardware. The architecture is built using React.js or Vue.js for the frontend and Node.js or Django for the backend, incorporating adaptive algorithms to adjust difficulty based on real-time gaze patterns.
Expected Outcomes: The primary outcome is a classification of a child's visual processing and brain speed levels (Low, Medium, or High) to support personalized learning pathways. **Social and Commercial Impact:** By providing an accessible, bilingual diagnostic and intervention tool, the project supports **SDG 4: Quality Education** by ensuring equitable learning opportunities for neurodiverse children in Sri Lanka.
Keywords: Dyslexia, Eye-Tracking, Brain Processing Speed, Gamified Learning, Bilingual Education.

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4. LIST OF ABBREVIATIONS

Abbreviation	Full Form
ADLs	Activities of Daily Living
AI	Artificial Intelligence
BLE	Bluetooth Low Energy
CNN	Convolutional Neural Network
CDR	Clinical Dementia Rating
CPU	Central Processing Unit
ESP32	Espressif Systems 32-bit Microcontroller
ESP8266	Espressif Systems Wi-Fi Microcontroller
GPS	Global Positioning System
IoT	Internet of Things
LSTM	Long Short-Term Memory
ML	Machine Learning
RTLS	Real-Time Location System
UI	User Interface
UX	User Experience

5. INTRODUCTION

5.1 Background and Context

Dyslexia is a neurodevelopmental learning disorder that significantly impairs foundational cognitive skills such as reading, spelling, and phonetic processing. It represents a major educational challenge, with an estimated global prevalence of 5% to 10%. In Sri Lanka, however, insufficient local research has led to inconsistent prevalence data ranging from 1% to 33%. Unlike typical learning variations, dyslexia manifests through persistent struggles with phonological awareness, visual tracking, and sustained attention, which impacts a child's performance across all academic subjects. This progressive academic struggle makes regular monitoring essential, not only to assess current literacy levels but to identify specific cognitive deficits that hinder early development. Early identification and intervention are critical to improving the quality of education and the overall well-being of neurodiverse children.

Traditional methods for assessing dyslexia in Sri Lanka are primarily episodic and outcome-based, relying on periodic classroom tests or subjective teacher observations. While helpful, these strategies often fail to capture the subtle, real-time "online" processing difficulties that are the initial markers of the disorder. For instance, a student who frequently loses their place in a text, hesitates over specific phonemes, or displays erratic eye movements may be showing early indicators of dyslexia that are easily overlooked in standard clinical or classroom examinations. This gap highlights the urgent need for intelligent, unobtrusive, and data-driven technologies that can follow a student's cognitive actions in real-world contexts and offer objective evaluations of their processing speed and visual behavior.

The increased availability of web-based technology, affordable camera sensors, and Artificial Intelligence (AI) models offers a significant opportunity to address this dilemma. AI-driven monitoring has been successfully used in assistive technology to track cognitive patterns without requiring intrusive medical hardware. When paired with machine learning algorithms capable of recognizing gaze patterns and forecasting learning outcomes, standard web cameras can serve as the foundation for intelligent educational support systems. These systems promise not only to provide real-time feedback but also to generate clinically useful data that enables early identification, stage measurement, and personalized intervention planning.

5.2 Proposed Solution: Interactive Learning Tool for Dyslexia

Within this context, this project has been designed to provide a structured framework for daily cognitive monitoring and brain processing speed assessment. The fundamental purpose of this component is to acquire a comprehensive picture of the student's visual and cognitive engagement during literacy tasks. By integrating low-cost and widely available hardware—specifically standard web cameras used for eye-tracking—the system ensures that student behavior is tracked with minimal intrusion. Furthermore, by incorporating lightweight AI models capable of real-time gaze analysis, this module adds a layer of intelligent interpretation to raw behavioral data, allowing the system to classify cognitive actions such as word fixations, saccades, and regressions. These patterns serve as crucial indicators of a student's functional condition and are used to predict their processing speed (Low/Medium/High) when evaluated in conjunction with literacy data.

The eye-tracking-based reading behavior subsystem focuses on collecting gaze movement during word interaction. By utilizing web-camera-based eye tracking, the system records fixation duration and attention loss while children interact with letters and words. This allows the system to monitor whether the student is following typical reading paths or demonstrating strange behaviors, such as frequent regressions or prolonged fixations on simple stimuli. These minor departures from normal visual patterns provide early warning indicators of cognitive impairment or reading-specific processing delays.

Brain processing speed assessment is also critical, as it is a foundaitonal cognitive skill. To address this, the module integrates level-based cognitive games that measure visual reaction

time and attention switching. These games help calculate brain speed indicators such as response delay and gaze-based decision time. Unlike standard tests that only record correct or incorrect answers, our behavior-oriented approach delivers greater contextual information regarding the child's processing efficiency.

The strength of this component rests not only in its individual assessment capabilities but in its integration as part of the wider 4-in-1 gamified platform. The gaze data, interaction metrics, and brain speed indicators comprise a multi-modal dataset that captures multiple elements of the student's cognitive profile. This data is used to generate personalized learning pathways and real-time feedback, helping caregivers and teachers make better-informed decisions regarding the child's education.

5.3 Summary

In summary, this module underscores the critical need for continuous, objective, and real-time cognitive monitoring systems for dyslexia. By merging low-cost eye-tracking with AI-driven brain speed assessment, the suggested solution advances beyond the limitations of standard outcome-based examinations. It creates a proactive approach where tiny changes in visual processing and cognitive behavior can be noticed early, and interventions can be adopted immediately. Ultimately, this component aspires to improve student outcomes, minimize teacher burden, and contribute to a more individualized, data-driven approach to neurodiverse education in Sri Lanka.

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6. Background & Literature Review

6.1 Global and Local Context of Dyslexia

Dyslexia is recognized as a neurological learning disorder that impairs reading, spelling, and cognitive processing, with a global prevalence estimated between 5% and 10%. In the Sri Lankan context, data is significantly inconsistent, with research indicating prevalence rates as high as 33% due to a historical lack of localized research and standardized identification tools. Children with dyslexia often struggle with phonological awareness, visual tracking, and sustained attention, which creates a barrier to academic success since reading comprehension underpins the entire primary school curriculum.

6.2 Eye-Tracking as a Cognitive Diagnostic Tool

Recent academic shifts emphasize the use of eye-tracking technology to provide a non-invasive "window" into the real-time cognitive processes of a reader.

- **Reading Behavior:** Efficient reading is characterized by shorter fixation durations, longer saccades (eye jumps), and fewer regressions (looking back at text).
- **Predictive Power:** Shorter eye-movement times in word recognition and meaning access are strong predictors of higher reading comprehension performance.
- **Implicit Comprehension:** Eye-tracking measures are found to be stronger predictors of implicit reading comprehension—the type of reading most frequently used in classrooms—than conventional tests like vocabulary or rapid naming.

6.3 Patterns in Dyslexic vs. Typical Readers

The literature establishes distinct visual behavior patterns that differentiate dyslexic readers from their typical peers:

- **Cognitive Load:** Dyslexic readers generally exhibit longer fixation durations and shorter saccades, indicating a higher cognitive load and more effortful processing during word recognition.
- **Visual Complexity:** Research indicates that dyslexic learners struggle more with visually complex stimuli (such as pictographs) because they cannot easily break them down into familiar sub-lexical components.
- **Sub-lexical Processing:** Studies suggest that while visual support can aid learning, a deeper understanding of character structures (morphemic components) is more crucial for improving reading among dyslexic learners.

6.4 Analysis of Existing Systems and Limitations

- **International Solutions:** Platforms such as Lexercise and GraphoGame offer practice-based interventions focusing on phonological training. However, these are designed for English-speaking populations and lack cultural adaptation for Sri Lankan curricula.
- **Sri Lankan Solutions:** Local tools like ARUNALU and LEXIS have introduced mobile-based screening and Sinhala phonological training. While valuable, these systems focus exclusively on Sinhala, neglecting the bilingual reality of the Sri Lankan education system.
- **Technical Gaps:** Existing tools largely measure the *outcome* of reading (correct/incorrect answers) rather than the *process* (how the eyes move). There is a lack of low-cost, web-camera-based solutions that combine behavior analysis with cognitive game performance.

6.5 Theoretical Foundation

This research is grounded in the **E-Z Reader Model**, which predicts fixation durations and saccades based on lexical access speed , and **Kintsch's Construction-Integration Model**, which outlines how readers transition from literal analysis to the creation of a mental situational model. The project seeks to bridge the gap between these clinical models and real-world educational support through a bilingual, AI-driven assessment framework

7. Research Gap

Despite the advancements in educational technology for neurodiverse learners, the domain of dyslexia care in Sri Lanka—particularly regarding early identification and bilingual support—faces several critical challenges that current solutions do not address.

Table 7-1:comparison of existing monitoring systems

Existing Approach	Focus Area	Limitations in Dementia Context
Standardized Literacy Tests	Reading and spelling accuracy	Captures only the final outcome; fails to identify real-time cognitive processing struggles or "online" reading behaviors.
International EdTech (e.g., Lexercise)	English-centric phonological training	Lacks cultural and linguistic adaptation for the Sri Lankan curriculum and local languages.
Local Solutions (e.g., ARUNALU, LEXIS)	Sinhala-only reading support	Neglects the bilingual (Sinhala-English) reality of the Sri Lankan education system.
Clinical Eye-Tracking Research	Visual behavior analysis	Highly expensive, relies on specialized hardware (e.g., Tobii), and is generally restricted to lab settings rather than accessible school tools.

7.2 Identified Critical Gaps

Based on a thorough analysis of existing systems and literature, four primary research gaps have been identified:

- **Absence of Process-Based Assessment:** Most current platforms measure the *results* of reading but ignore the *underlying cognitive processing*. There is no low-cost tool in Sri Lanka that analyzes gaze patterns, such as fixation duration and regressions, to identify the root causes of reading failure.
- **Neglect of Executive Function Foundations:** Existing local tools emphasize literacy practice (symptoms) without addressing executive function deficits like brain processing speed and sustained attention, which are foundational for reading success.
- **Fragmentation of Bilingual Data:** There is a lack of integrated systems that capture behavioral and cognitive data across both Sinhala and English, preventing a holistic understanding of a child's functional condition in a bilingual education system.
- **Limited Accessibility of Diagnostic Technology:** Advanced diagnostic methods like eye-tracking are currently unavailable to the general public in Sri Lanka due to hardware costs and the lack of user-friendly, web-based implementation.

The proposed system addresses these gaps by merging **low-cost, web-camera-based eye tracking** with **AI-driven brain-speed assessment**. By shifting from an outcome-based to a behavior-oriented analysis, the system identifies processing delays and cognitive markers that traditional clinical evaluations overlook.

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8. Research Problem

Dyslexia is one of the most urgent educational challenges in Sri Lanka, characterized by a persistent impairment in memory, phonological awareness, and reading fluency . Early detection and regular monitoring are critical for preventing academic failure and enhancing the long-term quality of life for neurodiverse children . However, traditional techniques for assessing dyslexia are generally limited to outcome-based evaluations, such as periodic classroom tests or subjective teacher observations conducted in controlled conditions .

While essential, these outcome-based examinations fail to capture the child's real-world functional abilities and the subtle, "online" behavioral changes that occur during the reading process itself. As a result, teachers and healthcare personnel lack ongoing, data-driven insights into the child's actual cognitive processing, making it impossible to intervene at the correct time or track progress precisely .

Current technical options, such as international software or basic mobile apps, provide only partial assistance and suffer from constraints including a lack of bilingual support (Sinhala-English) or a narrow focus on isolated literacy symptoms like letter recognition . Very few systems in Sri Lanka attempt to combine real-time behavior tracking with an integrated measurement of cognitive brain speed. This gap leaves parents and educators without a holistic tool to understand both the behavioral context and the underlying cognitive development of the disorder . Furthermore, advanced diagnostic tools like eye-trackers are often inaccessible due to high costs and technical complexity.

The research difficulty, therefore, lies in the unavailability of a complete, low-cost, and technology-driven system that can continually monitor the reading behavior of dyslexic children while simultaneously evaluating their brain processing speed. Addressing this challenge requires an innovative framework that bridges the gap between clinical evaluation and real-life monitoring, enabling targeted therapies and boosting early detection in a sustainable, bilingual manner.

9. OBJECTIVES

The proposed research project, **Interactive Learning Tool for Dyslexic Kids**, intends to address the critical need for improved, AI-driven solutions to support dyslexia care and management. By merging low-cost eye-tracking technology with gamified cognitive training, the system intends to track reading behaviors, recognize cognitive patterns, and quantify the processing speed of neurodiverse children in a non-intrusive and continuous manner.

9.1 Main Objective

To design and develop a bilingual (Sinhala-English), AI-driven web-based platform that enables continuous daily reading behavior monitoring and precise brain processing speed assessment, providing caregivers and healthcare professionals with real-time insights to enhance student safety, improve quality of care, and assist prompt interventions.

9.2 Specific Objectives (Individual Component: Khan N.A.B.)

- **Design Eye-Tracking-Enabled Reading Tasks:** Create a series of level-based literacy tasks designed to capture gaze behavior while children interact with letters, words, and short visual reading stimuli.
- **Integrate Web-Camera–Based Eye Tracking:** Implement a software framework that utilizes standard web cameras to record fixation duration, gaze shifts, regressions, and attention loss during reading and game activities.
- **Develop Brain Processing Speed Algorithms:** Implement algorithms to calculate brain speed indicators such as reaction time, response delay, and gaze-based decision time across different game levels.
- **Classify Visual Processing Levels:** Utilize cognitive indicators extracted from monitored data to classify children's visual processing and brain speed levels into three categories: Low, Medium, and High.
- **Implement Adaptive Learning Support:** Integrate a mechanism where brain speed and eye-tracking insights dynamically inform task difficulty and visual complexity, enabling personalized learning pathways.
- **Develop a Teacher/Caregiver Analytics Dashboard:** Create a secure interface that provides real-time notifications, heat maps of reading patterns, and structured reports to support early intervention decisions

- 10.Methodology

The research methodology for the **Eye-Tracking–Based Reading Behaviour and Brain Processing Speed Assessment** module follows a Design Science Research approach. It integrates real-time ocular tracking with gamified cognitive tasks to analyze the functional reading condition of dyslexic children . The process involves data collection from web-based sensors, AI-driven behavior analysis, and adaptive difficulty adjustment.

10.1 System Architecture and Data Flow

The system is built on a modular architecture that integrates hardware inputs with an intelligent processing layer:

- **Input Layer:** Captures real-time gameplay interactions and a continuous video feed from a standard web camera .
- **Processing Layer:** The **Eye-Tracking Engine** uses signal processing to filter raw gaze data and identify specific reading markers.
- **Analytics Layer:** Features are extracted to calculate brain speed indicators, which are then mapped to dementia/dyslexia stages using machine learning models .
- **Output Layer:** Generates heat maps of reading patterns and provides personalized learning recommendations.

10.2 Technical Methods and Algorithms

The module utilizes a combination of computer vision and statistical modeling:

- **Gaze Tracking & Calibration:** A five-point calibration process is used to align the web camera with the student's eye movements. The system calculates the **Received Signal Strength** of visual focus to determine the student's real-time position on the text with "room-level" accuracy of the character being read.
- **Ocular Metric Calculation:** Algorithms are implemented to differentiate between normal and abnormal activity patterns by measuring:
    - **Fixation Duration:** The time spent stationary on a word to access its meaning.
    - **Saccade Amplitude:** The length of jumps between different parts of the text.
    - **Regression Count:** The frequency of the eyes returning to a previously read word to reread it.
- **Brain Speed Assessment:** A lightweight **Convolutional Neural Network (CNN),** such as MobileNet, is used to classify behaviors in real-time. Reaction times and gaze-based decision times are evaluated against model confidence criteria to ensure reliable categorization.

10.3 Tools and Platforms

- **Frontend Development:** Built using **React.js** or **Vue.js** for a responsive, child-friendly user interface.
- **Eye-Tracking Framework:** Utilizes low-cost, web-based tracking libraries (e.g., **WebGazer.js** or similar) connected to standard 60 Hz web cameras to ensure the solution is practical for Sri Lankan schools.
- **Backend & Data Storage:** Employs **Node.js** or **Django** with a secure cloud-based database for time-stamped behavior logging.

10.4 Validation and Ethical Considerations

- **Validation:** System accuracy will be tested through pilot studies measuring location accuracy and classification precision . Metrics such as a $\geq 85\%$ classification accuracy for activity types will be used as acceptance criteria.
- **Privacy:** To address ethical concerns, video processing is performed locally on the user's device whenever possible to ensure sensitive data remains private.
- **Ethical Clearance:** The project requires formal approval from the **SLIIT Ethics Review Committee**, involving written parental consent and child assent before any data collection

11.REQUIREMENTS

This section defines the essential functions and constraints of the **Combined Eye-Tracking and Brain-Speed Assessment** module, ensuring it meets the needs of all key stakeholders.

11.1 Stakeholder Identification

- **Primary Users (Dyslexic Children):** Children aged 5–8 years who interact with the gamified tasks.
- **Secondary Users (Teachers & Special Educators):** Professionals in Sri Lankan schools who use the analytics dashboard to monitor progress and adjust classroom interventions.
- **Tertiary Users (Parents/Guardians):** Caregivers who require home-based reports to track their child's cognitive development.
- **Administrative Users (System Administrators):** Personnel responsible for maintaining the cloud database and ensuring data security compliance.

11.2 Functional Requirements

The functional requirements describe the specific actions the system must perform to analyze reading behavior and assess brain speed.

1. **Calibration Management:** The system shall provide a five-point visual calibration routine to align the web camera with the user's gaze before starting any assessment task.
2. **Gaze Data Acquisition:** The system shall record real-time eye-movement coordinates (x, y) at a minimum frequency of 60 Hz using a standard web camera.
3. **Reading Marker Identification:** The system shall utilize algorithms to identify specific reading behaviors, including fixation durations, saccade amplitudes, and regression counts .
4. **Brain Speed Calculation:** The system shall calculate cognitive indicators, such as reaction time and gaze-based decision time, during interactive gameplay.
5. **Performance Classification:** The system shall classify a child's visual processing level into "Low," "Medium," or "High" categories based on combined ocular and game performance data.
6. **Adaptive Difficulty Adjustment:** The system shall dynamically adjust task complexity (e.g., font size, visual clutter, or distractor frequency) based on the child's real-time gaze patterns.
7. **Heat Map Generation:** The system shall generate visual heat maps representing the areas of text where the child focused for the longest durations .
8. **Report Generation:** The system shall produce weekly progress reports summarizing accuracy trends, improvement trajectories, and identified weakness areas for parents and teachers.

11.3 Non-Functional Requirements

These requirements define the quality attributes and constraints of the system.

- **Usability:** The interface must be child-friendly, utilizing high-contrast visuals and audio scaffolding to support users with severe reading difficulties.
- **Performance:** The eye-tracking engine must process gaze data with minimal latency (under 100ms) to ensure the adaptive difficulty adjustment feels seamless to the user.
- **Privacy & Security:** All captured video data for eye-tracking must be processed locally on the device to prevent sensitive images from being stored on the cloud, addressing critical ethical concerns.
- **Accessibility:** The system must be web-based to ensure it is accessible on standard laptops or computers in Sri Lankan schools without requiring specialized hardware.
- **Reliability:** The system shall maintain at least 85% accuracy in classifying reading behaviors under varying lighting conditions typical of classroom environments

12.GANTT CHART

The Gantt chart of the development process is depicted as follows.

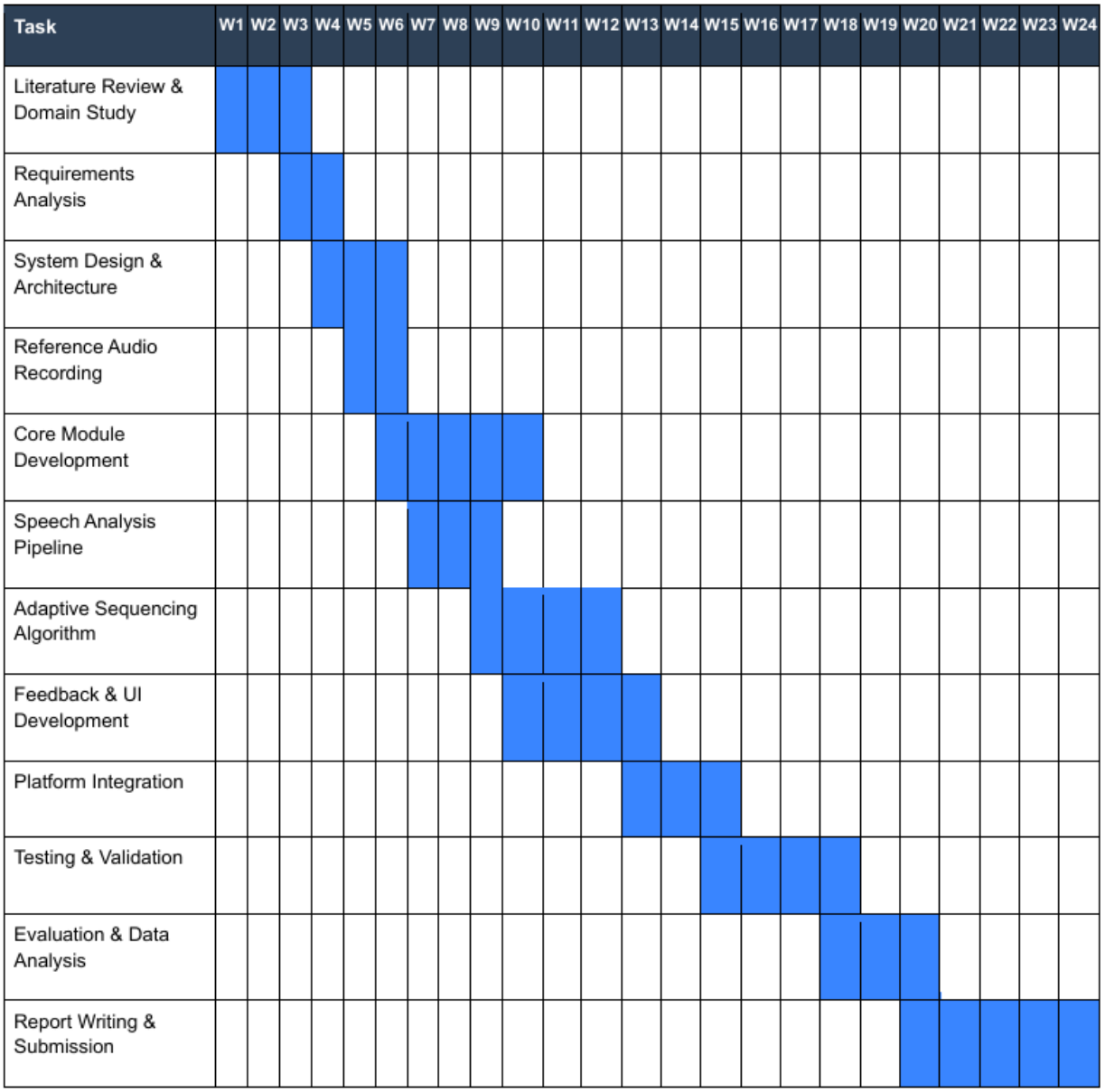


Figure 12-1:Gantt Chart

13.Work Breakdown Structure

Phase	Task	Sub-Tasks	Duration
1	Literature Review & Domain Study	Review dyslexia, phonological processing, SLI literature; study Sri Lankan context	Weeks 1-3
2	Requirements Analysis	Stakeholder interviews, functional & non-functional requirements definition	Weeks 3-4
3	System Design	Architecture design, UI wireframes, database schema, speech pipeline design	Weeks 4-6
4	Core Module Development	Implement listen-repeat-compare engine, phoneme capture, speech analysis pipeline	Weeks 6-10
5	Feedback & Adaptive Logic	Build visual/audio feedback UI, implement error-pattern adaptive sequencing	Weeks 10-13
6	Integration	Integrate with central dashboard, connect to analytics and progress tracking	Weeks 13-15
7	Testing & Validation	Unit testing, user testing with school children, usability evaluation	Weeks 15-18
8	Evaluation & Report Writing	Data analysis, SUS evaluation, dissertation writing and submission	Weeks 18-22

Figure 13-1:Work Breakdown Structure

14.Budgetary Plan

Table 14-1:Budgetary plan

Component	Estimated Cost (Rs.)
Hardware costs	20000/-
Software licenses	10000/-
Cloud services	7500
App Hosting on Google App Store	7475/-
Human effort cost	5000/-
Total Cost Estimated	**46475/-**

15.Commercialization

The commercialization strategy for the **D-Care Eye-Tracking and Brain-Speed Assessment** module focuses on bridging the gap between clinical neuropsychological research and the practical needs of the Sri Lankan education system.

15.1 Target Market and Customer Persona

- **Target Market:** Primary schools (public and private), special education centers, and speech-language therapy clinics in Sri Lanka.
- **Customer Persona: Primary:** Educational institutions seeking to provide inclusive learning environments for neurodiverse children.

  Secondary: Parents seeking home-based, low-cost cognitive support tools for children aged 5-8.

15.2 Value Proposition

- **Bilingual Accessibility:** Unlike international English-only solutions, this platform supports both Sinhala and English, reflecting the local educational reality .
- **Low-Cost Diagnostic Support:** By replacing expensive medical hardware with standard web-camera eye tracking, the system provides high-level behavioral data at a fraction of the cost.
- **Process-Oriented Insights:** Moves beyond "correct/incorrect" scores to provide teachers with heat maps and brain-speed indicators that identify *why* a child is struggling.

15.3 Revenue Model

- **B2B Subscription (Schools):** Annual licensing fee for schools to access the full 4-in-1 platform, teacher dashboards, and classroom-wide analytics.
- **Freemium Model (Parents):** A free version for basic screening, with a "Premium" tier for detailed progress reports and personalized learning recommendations.
- **Governmental Partnerships:** Collaborating with the Ministry of Education for large-scale implementation as part of national inclusive education initiatives.

15.4 Pricing Strategy

- **Penetration Pricing:** Initially offering low-cost entry for local schools to build a user base and generate "research-quality" datasets for system refinement.

- **Tiered Pricing:** Differentiation based on the number of students per institution and the level of data granularity required for reports.

15.5 Competitive Advantage

- **Unique Feature Integration:** The first tool in Sri Lanka to combine web-based eye tracking with cognitive brain-speed assessment.
- **Research-Driven Personalization:** The use of adaptive algorithms that dynamically adjust difficulty based on real-time gaze patterns rather than static test scores.
- **Data Privacy:** Local processing of video data ensures ethical compliance and high levels of trust among local caregivers and institutions.

15.6 Intellectual Property (IP) Considerations

- **Software Copyright:** Protecting the unique source code for the gaze-tracking integration and the adaptive difficulty adjustment algorithms.
- **Trademarking:** Securing the "D-Care" brand identity and system names.
- **Database Rights:** Proprietary ownership of the anonymized research datasets collected for system training and validation.

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